

ESTIMATION TRACK

VALIDATION TRACK

$\hat{\varphi}_\theta = \arg \min_{\varphi} \mathcal{N}(I \circ \varphi) + \alpha \mathcal{R}(\varphi)$
data term does not identify the map; the prior selects
tracking - registration - FE/biomechanics - supervised
- self-supervised - compact basis

$\theta \mapsto \theta$ $\theta \mapsto \theta + a \sin \theta$

$\lambda \theta = 1 + a \cos \theta$

Dice = 1, HD = 0

7 Ecc

ES

Specifications 01–08 together define one regional estimand;
 ○ readouts and 08 (6 segments) are computed from the curves

R representation — project the known field onto the basis

Mask, landmark, registration or inter-method agreement 22
 Material-sensitive reference (known motion, DENSE, tagging); no focal deficit 14
 Downstream label: class, LGE, outcome or expert reading 8
 Validation data not reported 3
 Prescribed focal deficit with known motion (independent) 1

0 5 10 15 20 25
 entries (n = 48; manuscript Fig 3)

M: magnitude · L: location · E: extent · T: timing, Ecc/Err/ErrL: circumferential/radial/longitudinal strain, Ecs: end-systole, PSS: post-systolic shortening, FT: feature tracking, GL: Green-Lagrange, HD: Hausdorff distance, LoA: limit of agreement, WMA: wall-motion abnormality